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THE ROTATION OF IBERIA; CAENOZOIC PALAEOMAGNETISM FROM PORTUGAL *

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ABSTRACT

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A palaeomagnetic study of the Lisbon volcanics (Eocene) has revealed a variety of magnetization directions suggesting long-lasting remagnetization processes. The entirely dominating magnetic constituent has a Curie temperature of about 350°C, but this phase undergoes a spontaneous break-down when being heated to temperatures slightly above 400°C. There are strong reasons for believing that the original titanomagnetites have become nearly completely oxidized to cation-deficient phases (titanomaghemites); i.e., the original thermoremanent magnetization is likely to have become extensively replaced by low-temperature chemical magnetizations. Despite severe experimental problems in deducing the various magnetization directions of the record the data tend to support some interesting geophysical deductions. First of all, the rotation of Iberia does not appear to have started at the time the Lisbon volcanics were laid down. A re-interpretation of marine geological data from the Bay of Biscay is compatible with this conclusion in that an early Oligocene age of the Biscay abyssal basement is inferred. This would place the creation of the Bay of Biscay fairly simultaneously in time with that of the Western Mediterranean basin. Relatively soon after the opening of the Bay of Biscay a marked polar migration towards the present position appears to have taken place. The timing of this latter event may be Late Oligocene/Early Miocene. The reason for the discrepancy between the present palaeomagnetic results from the Lisbon lavas and those of previous studies is dealt with in some detail.

INTRODUCTION

Based on a study of post-Palaeozoic crustal deformations in southwestern Europe, Carey (1958) suggested that the origin of the Bay of Biscay was coupled to an anticlockwise rotation of Iberia of about 35°. Later, this idea has gained strong confirmation from palaeomagnetism (Schwarz, 1963; Van Dongen, 1967; Van der Voo, 1967, 1969). Also, Bullard et al. (1965) had to close up the Bay of Biscay (at the 500 fathom depth curve) in order to obtain the best "least-squares fit" of the continents bordering the North Atlantic (i.e. the suggested predrift configuration). Recently, marine geophysics has provided information about the crustal structure in the Bay. Both seismic and gravity data (Bacon et al., 1969) suggest a typical oceanic crust, and the observation of linear magnet-

* Norwegian geotraverse contribution no. 58.

ic anomalies (Matthews and Williams, 1968; Le Borgne et al., 1971) above the abyssal part of this sea gives further support in the same direction. A reliable time estimate of the opening would be of considerable geophysical interest through its bearing on the structural history of the Northeast Atlantic and possibly also of the Western Mediterranean Sea. Based on sea-floor spreading data Williams and McKenzie (1971) suggest an Upper Jurassic/Lower Cretaceous opening while Le Pichon (1968) favours a Middle/Upper Cretaceous age for this megatectonic event. However, these age estimates are rather tentative and it was thought therefore that the Lisbon volcanics which are of Eocene age might contain further palaeomagnetic information to this question. The formation has previously been subjected to palaeomagnetic studies (Watkins and Richardson, 1968a; Van der Voo, 1969; Van der Voo and Zijderveld, 1971) but the geophysical interpretation of the data has become a matter of debate (Van der Voo, 1968; Watkins and Richardson, 1968b; Storetvedt, 1970; Watkins and Richardson, 1971). Based on the experimental results of Watkins and Richardson (1968a), Storetvedt (1970) has argued that the fossil magnetization in question is likely to have a much more complex build-up than suggested by the previous authors. In fact, the results of the present study strongly confirm this suggestion in that chemical magnetization acquired at ambient temperatures, involving times of field reversals as well as relative polar wandering, appears to have extensively replaced the original magnetic record. Despite the complex magnetic structure, however, important suggestions concerning Caenozoic palaeomagnetism and the time of opening of the Bay of Biscay can be deduced.

GEOLOGY AND SAMPLING

The Lisbon volcanics are stratigraphically situated between Upper Cenomaniam carbonates and the Oligocene Benfica Formation; the interbasaltic layers containing Eocene freshwater molluscs, vertebrates and plants (Zbyszewski, 1964). The lava sequence has been involved in a mild phase of tectonic disturbances which included Upper Mesozoic as well as Oligocene strata. A total of thirty hand samples have been collected from seven lavas. Five of these flows were practically horizontal, whereas the remaining two (flows 2 and 3), which were collected within the Loures fold structure northwest of Lisbon, appear to have a northerly dip of around 40°. The sample numbering relative to the different flows is as follows:

lava 1 – samples 1– 5; lava 2 – samples 6–10;
lava 3 – samples 11–12; lava 4 – samples 13–14;
lava 5 – samples 15–20; lava 6 – samples 21–26;
lava 7 – samples 27–30.

A magnetic orientation method was applied as the weather did not permit the use of the sun compass. Consequently, all data have been corrected for the present local magnetic declination of 9.5°W. This procedure may introduce a random declination error in the results. According to Watkins and Richardson (1968a), however, such deflections are likely to be less than 10° in the present case. As will be seen below, problems of this kind are entirely subordinate compared with the experimental problems associated with the complex magnetic history of this formation.

LABORATORY METHODS

In the laboratory up to six 3/4 inch cylindrical specimens were cut from each sample. The N.R.M. of these specimens was measured on various astatic magnetometers. The N.R.M.-analysis was done by means of progressive application of increasing alternating magnetic fields ranging upwards to 1050 Oe. Steps of 30–60 Oe were used in the study of the low-stability fraction of the remanence, while steps of 160 Oe or more were applied in the analysis of the more stable magnetic components.

By means of a translation balance, thermomagnetic cycling experiments up to 700°C have been carried out on at least one specimen from each sample; a complete heating cycle lasting for 25–50 minutes. In the most rapidly heated specimens there is a maximum thermal lag of 15°C. The magnetic field applied varied between 1.8 kG and 6.6 kG depending on the amount of strongly magnetic material present in the specimens. The heatings were done in air. In order to study any oxidation effects both finely powdered specimens and "whole rock" (chips) specimens have been used. The Curie-point (T_c) estimates are taken as the point of maximum downward curvature of the $J_s(T)$-curves. In some cases where the $J_s(T)$-curves decay linearly to zero, the point at which the magnetization disappears is taken as the T_c-estimate.

MAGNETIC MINERALOGY; $J_s(T)$-DATA

Characteristic examples of $J_s(T)$-curves are shown in Fig. 1. The Curie-point data together with other properties obtained from the $J_s(T)$-measurements may be summarized as follows:

Heating curves

(1) A very sharp J_s-reduction takes place in general at intermediate temperatures; the vast majority of these discontinuities occurring at about 350°C. The $J_s(T)$-curves are reversible below approximately 400°C, but irreversible above this temperature. An accurate T_c-determination is sometimes disturbed by a sudden increase in the resultant spontaneous magnetization which initiates at around this temperature (see below). The 350°C-phase has a tendency towards being better marked in the chip specimens.

(2) An increase in the total saturation magnetization within the temperature interval 380–500°C occurs in approximately 35% of the thermomagnetic heating curves. The same tendency is recognized in many more cases in that the total J_s, instead of decreasing, frequently stays constant through a few tens of degrees within this interval. The magnetization increase is more pronounced in the chip specimens.

(3) With two exceptions high-temperature Curie points are always recorded, but except for lava 1 the high-temperature T_c-phase carries in general less than 20% of the total saturation magnetization. Powdered samples may have multiple Curie temperatures in the range 500–700°C while the chip specimens exhibit single high-temperature Curie points in the range 500–580°C.

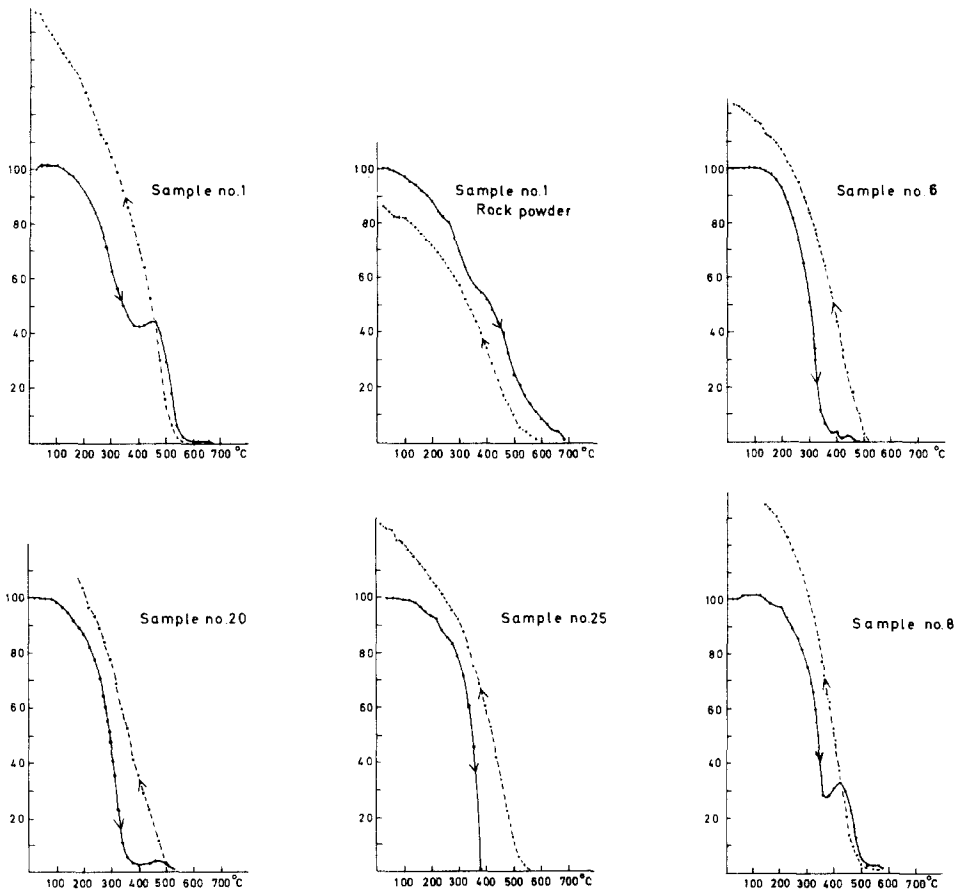


Fig. 1. Representative examples of saturation magnetization (arbitrary scale) versus temperature. Note the difference between a chip specimen and a powdered specimen in sample 1.

Cooling curves

The cooling curves are nearly always single-phased with T_c strongly peaked around 520–540°C. Some powdered specimens, however, have Curie temperatures in the range 600–700°C. The terminal T_c-concentration around 530°C is probably due to nearly pure magnetite; the T_c-suppression below 570°C being possibly caused by impurity cations such as Mg and Al (Löffler et al., 1965).

There is no doubt that an evaluation of the 350°C-phase, which represents the most important magnetic constituent of these rocks, is of prime importance for a proper interpretation of the remanence data. There are several reasons why this component is unlikely to be a virgin state titanomagnetite. First of all, a T_c of about 350°C is somewhat high for original basaltic titanomagnetites. The Curie-point values for stoichiometric titanomagnetites are considered to be in the 0–300°C-range (Carmichael and Nicholls, 1967)

with a peak in the interval 100–200°C (Creer and Petersen, 1969; Ade-Hall et al., 1971). Furthermore, the rather spontaneous disappearance of the 350°C-phase when the rock is heated to temperatures around 400°C does not suggest that the process of oxidation of unaltered titanomagnetites takes place. Thus, Creer and Petersen (1969) have reported longtime isothermal heating experiments on stoichiometric titanomagnetites at temperatures of 400°C and 520°C respectively in order to produce single-phased $J_s(T)$-curves which in the present study (at temperatures between 400 and 500°C) have a spontaneous appearance. The spontaneous intensity increase between approximately 400 and 500°C is most likely caused by an unmixing of the 350°C-phase into a number of new phases (Readman and O'Reilly, 1971) but with a Fe-rich component with T_c around 550°C (and with a stronger specific magnetization than its parent phase) as the important magnetic product.

Following the idea of a spontaneous break-down, the 350°C-phase is likely to be a cation deficient titanomagnetite (titanomaghemite). The most important magnetic component resulting from the postulated unmixing appears to be pure magnetite. If powdered samples are used the effect of the increased oxidation potential (the reaction rate is proportional to the area in contact with an oxidizing environment) is to obscure the 400–500°C transitions in addition to producing a high-temperature phase with T_c in the interval 600–700°C (probably a chemically stabilized cation deficient spinel).

The Curie-point data obtained from the heating curves of this study broadly confirm the limited T_c-data given by Watkins and Richardson (1968a). The basic difference, however, is that the 350°C-phase, which is such a predominant feature in the present analysis, seems to be much less pronounced in the previous study. The most likely explanation is that powdered specimens combined with a too slow heating rate has been employed by the previous workers (their report of T_c-estimates well above 600°C for deuteric oxidation classes I and II tend to strengthen this hypothesis).

An interesting result given by Watkins and Richardson is that a specimen in which the magnetic mineral content was so strongly dominated by maghemite that no deuteric oxidation class could be assigned to it, gave a single T_c of 375°C. This result is in complete agreement with the present interpretation of the 350°C-component.

It has been shown (Watkins and Richardson, 1968a) that the deuteric oxidation classes I–IV are present in the Lisbon lavas; i.e., specimens with dominantly optically homogeneous "titanomagnetite" grains may occur along with specimens in which the magnetic grains have become strongly subdivided by rhombohedral lamellae development. The unexsolved grains would be very unstable to subsequent oxidation and the present experiments suggest that the 350°C-maghemite phase is likely to have practically completely replaced the original stoichiometric titanomagnetite. A question of ultimate significance for an adequate palaeomagnetic interpretation is the importance of the postulated maghemite component relative to any remaining deuterically exsolved magnetite. Except for two specimens, where nearly pure magnetite seems to be the only magnetic constituent, maghemite contributes between 40 and 100% of the total J_s and there may be great variations within a single flow. This indicates that the remanent magnetization of the Lisbon volcanics is likely to be strongly governed by components formed by oxidation at ambient temperatures.

REMANENCE PROPERTIES

The majority of the specimens had N.R.M. intensities varying between approximately $2 \cdot 10^{-3}$ emu/cm^3 and $4 \cdot 10^{-2}$ emu/cm^3, the overall average being $1.32 \cdot 10^{-3}$ emu/cm^3.

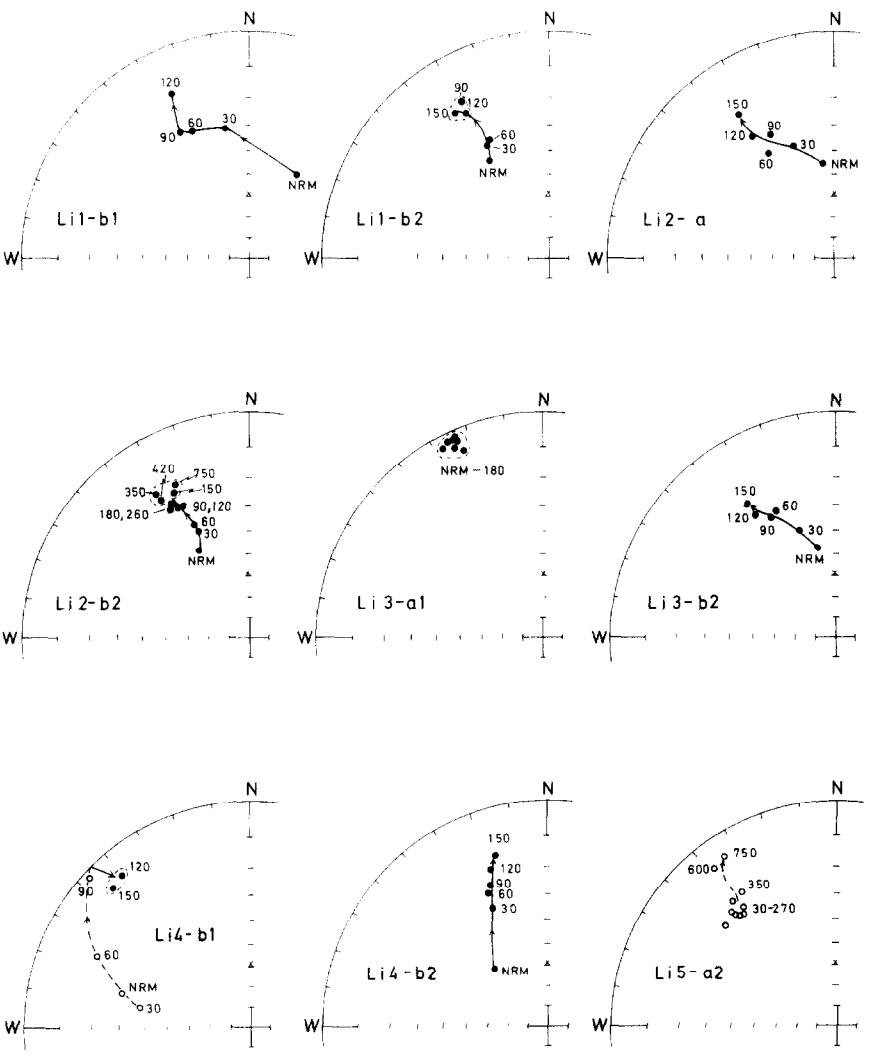


Fig. 2.

Fig. 2–6. Examples of directional behaviour versus progressive alternating field demagnetization. Directions marked T.C. have been subjected to tectonic corrections. Full circles represent downward inclinations whereas open circles are upward seeking magnetism. See text for further details.

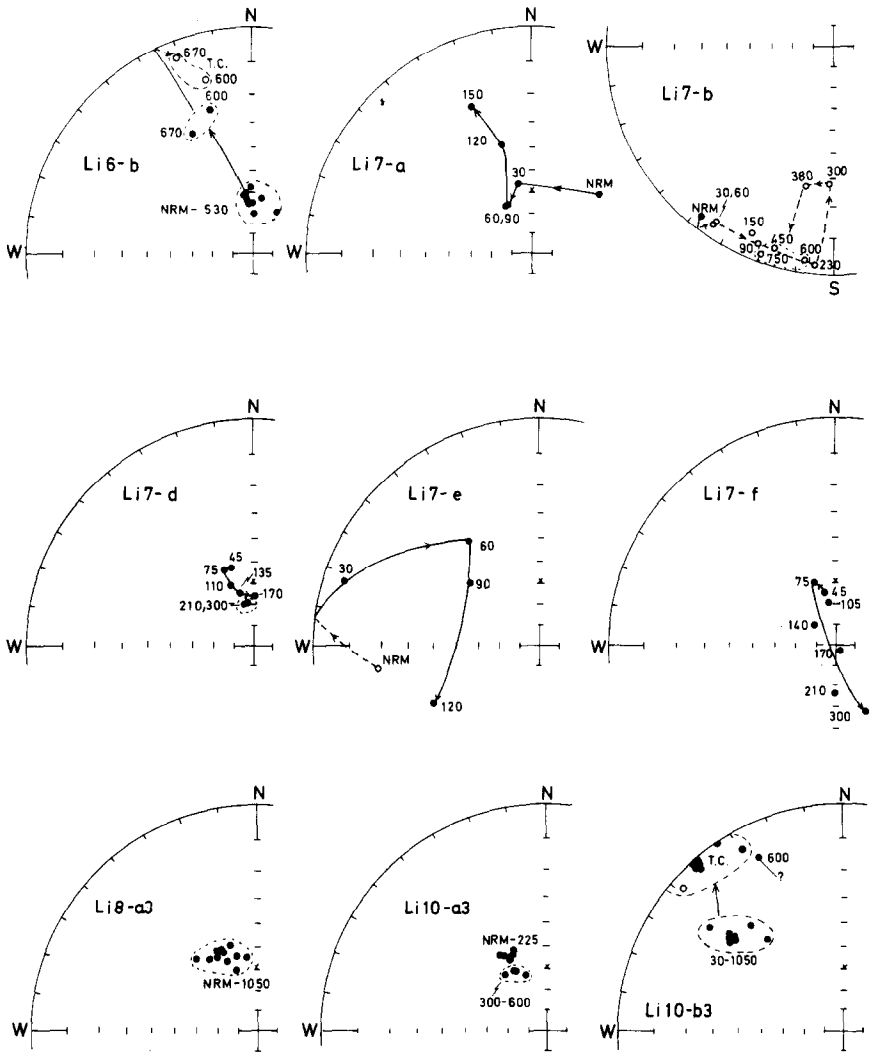


Fig.3.

In correspondance with the previous results, the N.R.M. intensity normally displays a rapid decay when subjected to increasing alternating fields. A major problem in this study is that the remanence vector of most specimens becomes indeterminable at some stage of a.c.-analysis. According to the measuring results the problem appears to be caused by induced magnetization (possibly combined with an increase of unstable magnetism during demagnetization) becoming the dominant component after a major reduction of the remanence intensity. The problem is particularly pronounced in specimens of low magnetic stability in which the remanence parameters may be masked after demagnetization to 150 Oe or so. On the other hand, in a number of magnetically harder rocks the remanence

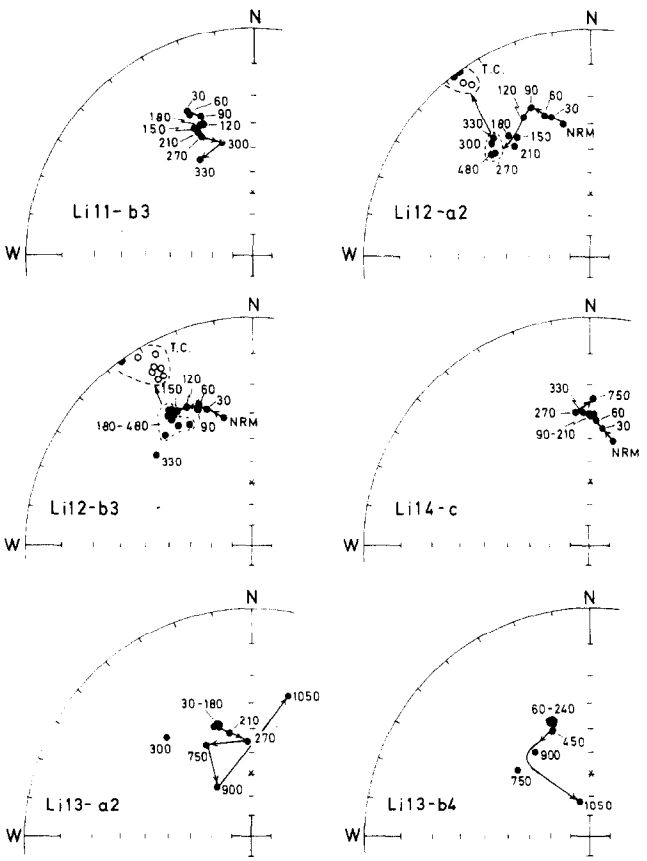


Fig.4.

could be reliably measured after a.c.-treatment in considerably higher fields.

The fairly complex magnetization build-up of this formation is, in general, well demonstrated within a separate lava flow, but the directional diversity may as well be revealed within a single hand sample (cf. for instance specimens from sample no. 7 of Fig.3). As a general rule the original magnetization vectors have moderately-steep downward inclinations with northerly declinations. Very few of the initial directions remain stable on progressive demagnetization. Figs.2–6 include the directional data obtained during the demagnetization process of 39 specimens spread over the entire collection. The number of examples constitutes approximately 50% of the total number of specimens tested, but includes nearly all those for which the remanence could be reliably measured after treatment in 300 Oe or more. The variation in maximum a.c.-field in the figures is due to the measuring problems as outlined above.

In addition to an unstable component, residing in low coercive force domains, the underlying "stable remanence" in six of the seven lavas studied appears to consist of a variety of magnetization vectors having similar magnetic stability on average. The general

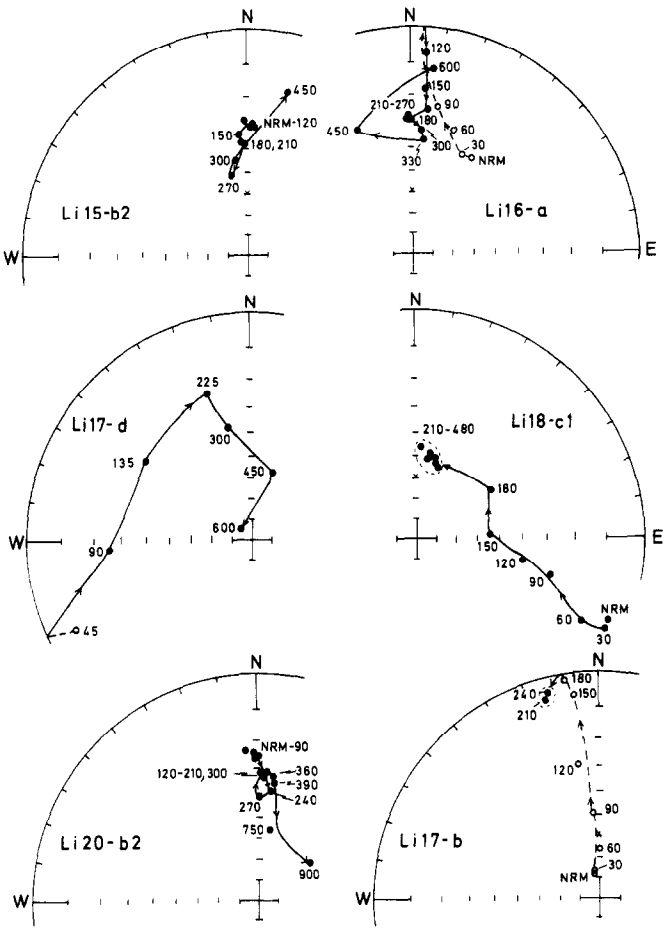


Fig.5.

directional behaviour when demagnetizing such a complex magnetic framework is systematic movements between two or more directions of the system. Unfortunately, these directional trends are very often disrupted in a random way before terminal directions have been achieved. On the other hand, a remagnetization product which generally shows up with a deviating net magnetization may well exhibit complete directional constancy in the stability range available for study. Consequently, magnetic stability alone is no measure for palaeomagnetic reliability. In practice it is only possible to determine the direction of the most stable component of a composite magnetization. In the present case, the successful verifications of such directions are very few and because of the experimental problems they are of highly variable quality. Despite the analytical difficulties, it seems possible to extract three different magnetic axes. These are based on the following remanence directions:

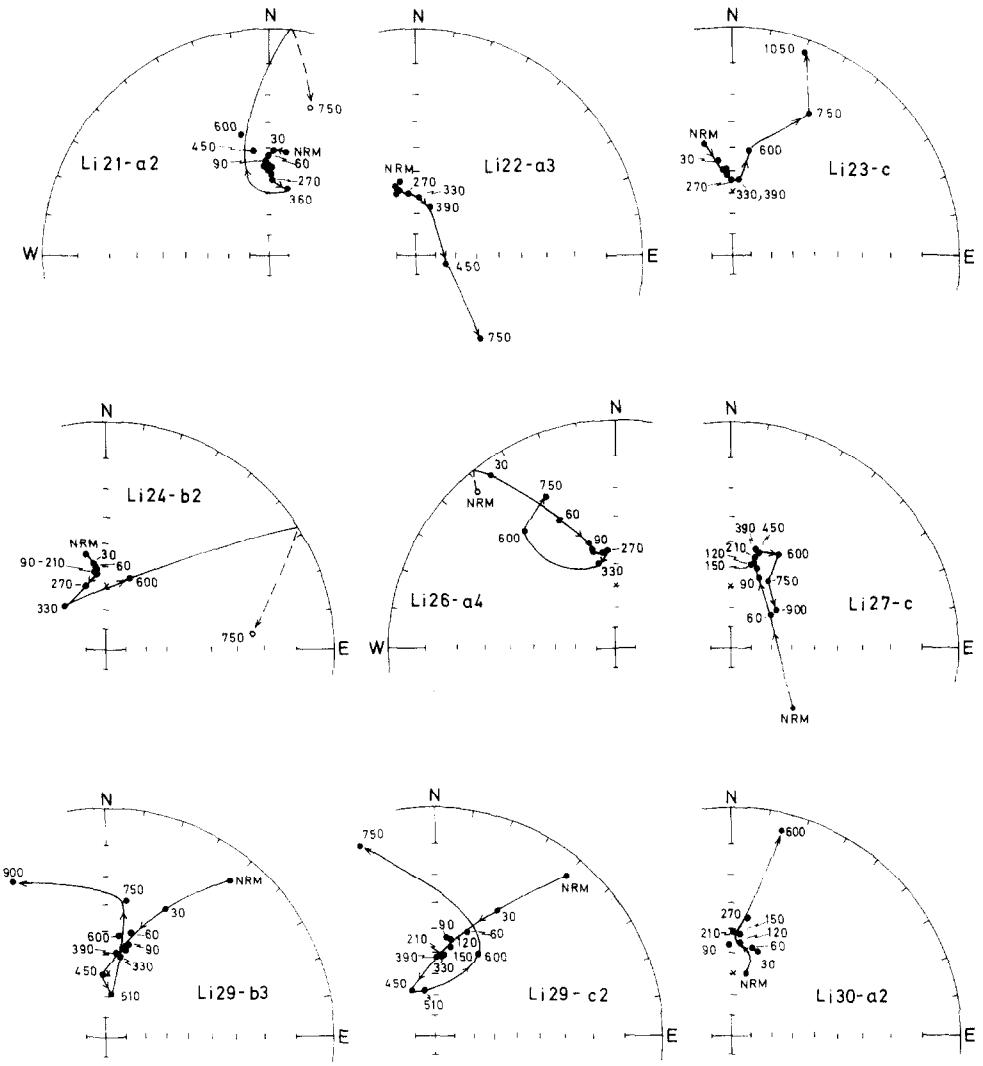


Fig.6.

(1) *A NNW-component with shallow inclination*

To a greater or lesser extent, the NNW-direction has been found in six of the seven lavas studied. It is particularly pronounced in lava 1 (Fig.2) where it apparently forms the only stable magnetization present. In other cases this component may be strongly overprinted by other stable magnetizations, with nearly identical coercive force spectra, so that only indications of its existence may appear (cf. 26-a4, 29-b3 and 29-c2 of Fig. 6). Despite the problems of experimentally verifying the terminal direction there is quite

strong evidence of a directional convergence in a position at approximately 330, +5. As will be discussed below, the NNW-component probably is the only magnetization of the considered record for which correction for tectonism is to be applied. In the present collection, however, this correction only appears to have practical importance in flows 2 and 3. Group I is thought to be defined by the mean of the encircled (and most stable) directions for each of the following 9 specimens: 1-b2, 2-b2, 3-a1 and 4-b1 of Fig.2; 6-b and 10-b3 of Fig.3; 12-a2 and 12-b3 of Fig.4; 17-b of Fig.5. The relevant data of Fig.3 and 4 are those having been corrected for tectonism (see later).

(2) *A shallow reversed component striking nearly due south*

This direction is determined in one specimen only, 7-b (Fig.3). Further evidence in support of this hypothetical component is provided by some data of Watkins and Richardson (1968a) in that three of their specimen directions (2-1, 5-3 and 11-1) tend to converge in this position after demagnetization to 400 Oe. At least some of the many specimens exhibiting southerly directional movements (in the present study as well as in that of Watkins and Richardson, 1968a) may have their most stable component in this shallow reverse position. Furthermore, some of the specimens for which the remanence vector is moving in a NNE direction (cf. 15-b2 and 16-a of Fig.5; 23-c and 30-a2 of Fig.6) may tend to establish the normal counterpart of direction II. Hence, the existence of a practically horizontal magnetization axis striking slightly east of true north is inferred.

(3) *A normal magnetization striking nearly due north and with a downward inclination of about 60°*

This direction is first of all thought to be defined by the mean of the encircled directions for the following four specimens: 7-d, 8-a3 and 10-a3 of Fig.3; 18-c1 of Fig.5. Furthermore, the mean remanence direction below approximately 600 Oe for specimen 6-b (Fig.3) has been included in this group. The acceptance of this latter specimen direction (of intermediate stability) as a palaeomagnetic component (in the general case this would not be possible) is based on the following experimental evidence: (a) the two components of specimen 6-b appear to have discretely different stability ranges; and (b) the most stable component (the NNW-direction) seems to be responsible for less than 4% of the total remanence intensity. The remanence direction of specimen 7-b performs an excursion towards a steeper negative inclination when being demagnetized in peak a.c.-fields between approximately 250 and 450 Oe. This behaviour possibly indicates the presence of a second reversed component antiparallel to direction III. Moreover, some specimens which apparently are in the course of changing remanence polarity by directional movements through shallow NNE–NE positions (cf. 21-a2 and 24-b2 of Fig.6) may well have their most stable component along such a steeply inclined reversed direction.

In the following, the magnetizations concerned are also referred to as axes I, II and III respectively.

THE PALAEMAGNETIC RECORD AND ITS INTERPRETATION

With reference to the foregoing section, there are only fifteen specimen directions in this study for which there is reasonable experimental evidence that palaeomagnetic direc-

tions have been obtained. These directions are shown in Fig.7. Some of them are experimentally well established, in that several high-field demagnetization steps define the terminal position whereas others are less reliable as certain magnetic properties (causing measuring problems) have prevented a satisfactory experimental determination of the stable end points. Nevertheless, the overall directional behaviour seems to be in harmony with the directions obtained.

There appears to be some connection between magnetic mineralogy and directional characteristics during demagnetization. Thus, lava 1 (Fig.2), in which the high Curie-point phase (~ 550°C) contributes around 50% of the total saturation magnetization, appears to have a single stable component: the NNW-direction. In the other lavas examined, where on the average the high Curie-point phase plays a subordinate role, a variety of stable components emerges. This might perhaps indicate that in lava 1 the stable remanence is dominantly of deuteric origin (T.R.M.) and carried by the 550°C-phase (magnetite?). The variety of stable components in the other lavas is compatible with a much greater predominance of the 350°C-phase which is considered to be a product of low-temperature oxidation. Unfortunately, the observations do not appear to allow such clear-cut explanations. Sample 13, for example, has practically completely reversible thermomagnetic curves with a T_c around 550°C so that one might expect to find a single stable component (of deuteric origin). Instead, the high-stability region of the remanence suggests a multi-component magnetization. This might well imply that not only original homogeneous titanomagnetite has been subjected to low-temperature oxidation, but that a certain cation-deficiency has been brought about in the deuterically exsolved magnetite as well. Whatever the explanation may be, the predominance of maghemite in this formation, coupled

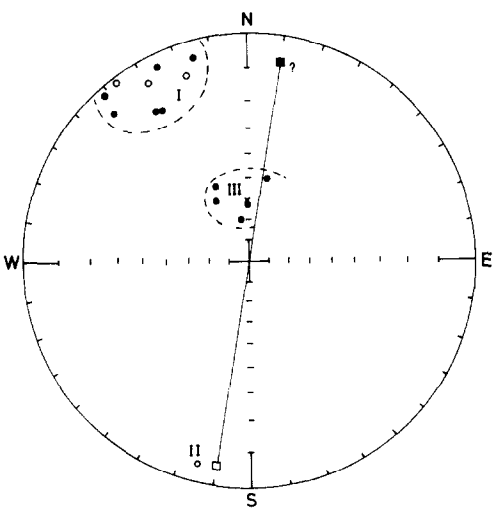


Fig.7. Acceptable palaeomagnetic results: specimen directions. Magnetization axis II is mainly based on evidence from directional trends. This axis tends to be in good agreement with that of the Early Tertiary for stable continental Europe. The plot convection is as for Fig.2–6. The cross represents the present dipole field direction.

to a variety of magnetization directions, strongly suggests that low-temperature chemical magnetization (perhaps combined with self-reversal mechanisms) contributes to the major fossil magnetization. The time taken for these processes can only be a matter of speculation but most likely they may have started already during the late stages of cooling and proceeded more or less continuously for a considerable time interval. The original magnetic structure has most likely suffered a practically complete break-down. This has obvious consequences for the application of some of the traditional concepts and procedures in palaeomagnetism. Based on the evidence available it would for example be a too naive approach in this study to consider the specimen direction of Fig.7 as representing spot readings of ancient fields. In fact each specimen direction is more likely to constitute a certain average field reading for a particular polarity epoch (or epochs of same polarity). The major directional scatter within a separate group of directions can probably be ascribed to minor contributions from the other equally stable directions (although a terminal direction has apparently been achieved on demagnetization there is no guarantee that this is a completely clean single component). Though the palaeomagnetic directions of this study are unevenly distributed between sites (lavas), the data given in Table I are based on unit weight for each specimen. According to the arguments presented this is considered as the most correct utilization of the observations.

TABLE I

Palaeomagnetic mean directions

Group	N	D	I	$\alpha 95$
I	9	330.2	+ 3.8	8.1
III	5	350.3	+58.8	11.8

Before entering into a discussion of the magnetic history of the Lisbon volcanics, it would be appropriate to recapitulate what is known the early Tertiary field for stable continental Europe. Unfortunately, the observations available so far appear to suffer from great uncertainties. Thus, a recent study of the Antrim basalts of Northern Ireland (Lövlie et al., 1972) suggest that the declination as well as the inclination of the fairly stable bulk magnetization of this formation have both been deflected by approximately 20° from the supposed true position of the original field. This conclusion has been reached by comparing the basaltic remanence behaviour on demagnetization with the remanence of interbasaltic laterites rich in haematite. An early Tertiary dipole field direction for Northern Ireland at about 180, –40 has been proposed (Løvlie et al., 1972). For the location of Lisbon this postulated field should be nearly horizontal and striking approximately due N; the north-seeking magnetism having a dip of around 10° below the horizontal (positive). In fact, magnetization axis II, which is weakly developed in this formation, seems to be in close agreement with the postulated early Tertiary field for extra-alpine Europe. Magnetization II should therefore be of post-rotation origin.

Magnetization axis I, however, differs from axis II by being rotated anticlockwise by around 30–40°. This postulated angle of rotation is the same as predicted by Bullard et

al. (1965) and by previous palaeomagnetic workers dealing with older rocks from the Iberian peninsula. Thus, axis I appears to have been acquired prior to the rotation of Iberia. Hence, palaeomagnetism of the Lisbon volcanics seems to suggest a Late- or post-Eocene opening of the Bay of Biscay. If the geomagnetic reversal frequency at the time of opening was approximately like that of Late Caenozoic times the opening process must have been completed within a few million years (Le Borgne et al., 1971). A relatively rapid opening, combined with polarity inversions and a slow rate of C.R.M.-aquisition, probably effectively destroyed the possibility of reasonably recording the geomagnetic field relative to Iberia during the act of rotation. On the other hand, the dipole field existing before rotation appears to have persisted through a sufficient period of time after this event to leave a measurable trace in the magnetic record. At some subsequent time a rapid dipole shift towards the present axis seems to have occurred. The rather feeble development of magnetization axis II (considered to have been imposed subsequent to rotation) tentatively suggest that the postulated axial shift does not seriously post-date the rotation. It must be stressed, however, that if this geomagnetic field (II) was characterized by rapid polarity fluctuations, the associated fossil magnetism could well be fairly indistinct even if the time interval concerned was of considerable length. It may be argued that group III represents magnetization acquired in recent times. However, considering the manner in which the Lisbon volcanics appear to have acquired their present remanence, together with the fact that the group III directions are as stable as any of the other directions estimated, there is no sensible reason to adopt the conventional interpretation, i.e., to suppose a Quaternary origin for all remanence directions which are close to that of the present field. The alternative explanation, which is favoured by the present author, is that the geomagnetic field for a certain age slot, immediately following the time of the inferred axial shift, had an excess of normal polarity. The existence of reversed magnetization corresponding to the final field axis is only tentatively inferred from the present experiments. The reason for this may be that the reversed magnetization is closely antiparallel to a more dominating normal component, both having similar magnetic stability. Under such circumstances the reversed contribution would at best show up with a certain southerly direction change during demagnetization (at a low intensity level). The feasibility of adding C.R.M.-components to the palaeomagnetic record will probably diminish as oxidation proceeds. In fact, the existing relative contribution of unstable and more stable magnetization components may to a large extent be the result of chemical processes having terminated millions of years before the Brunhes normal polarity epoch.

A recent consideration of marine geological data from the Bay of Biscay has given strong support to the present palaeomagnetic results in that a lowermost Oligocene age of the Biscay oceanic crust is suggested (Storetvedt, 1972).. Immediately after rotation the area appears to have been subjected to a strong northward compression resulting in an eastward displacement of the Iberian plate (of some 300–400 km) into its present position; the north Pyrenean fault and the "Azores–Gibraltar" fracture zone probably serving as guide lines for this movement. It appears reasonable to suppose that the fold structure in the Lisbon region developed as a result of this compression. Most likely both magnetization axes II and II postdate this deformation.

COMMENTS ON PREVIOUS RESULTS

The remanence behaviour versus progressive demagnetization as outlined above resembles closely that revealed by the data of table 5 of Watkins and Richardson (1968a). On the other hand, the latter authors obviously have based their analysis on too simplified concepts (Storetvedt, 1970) so that essential information about the magnetization build-up has passed unnoticed. In fact, only six or eight of their 39 specimens show reasonable directional stability. Nevertheless, these few directions are in fairly good agreement with either group I or group III of the present study. The magnetic complexity seen in these two investigations is in striking contrast to the occurrence of an apparent single high-stability direction recently obtained in another study of the considered lavas (Van der Voo and Zijderveld, 1971). The explanation for this discrepancy is likely to be sought in one of the most critical experimental factors in remagnetization problems: the specimen size. Thus, the acquisition of chemical remanence in a lava flow must be a function of micro-environmental oxidizing conditions which in turn is time-dependent, i.e., various parts of even a very small rock fragment may have become oxidized (and remagnetized) in distinctly different periods of time. Hence, small rock cylinders as those having been used by Watkins and Richardson (1968a) and by the present author are statistically in favour of a more uneven distribution of the various remanence components present (with respect to magnetic hardness as well as intensity of magnetization) than larger specimens (such as hand samples) which are used by Van der Voo and Zijderveld (1971). This means that while one may have a fair chance to unravel the magnetic history of a formation by analyzing small specimens, larger ones are much more likely to give a "single high stability direction" corresponding to the resultant of any existing multicomponent system. The same "average reading" will result when site mean directions are computed uncritically.

Returning to the formation under discussion, the present analysis has revealed two major stable components (I and III) of nearly equal importance. Also, they are both of normal polarity and of equal stability on average. By application of larger specimens as in the study of Van der Voo and Zijderveld one would expect to obtain stable remanence directions lying approximately midway between groups I and III. This is also what has been observed. However, the resultant directions do not retain exactly their true westward declination probably because tectonic corrections have been applied to a composite magnetization; there is every reason to suppose that only the shallow-inclined sub-component (group I) should be subjected to such an adjustment.

Van der Voo and Zijderveld claim to have isolated a single prefolding component in that a tighter grouping is obtained after applying the unfolding test in particular to the fold structure northwest of Lisbon. As has been pointed out by the present author several times, the fold test has a severe limitation in that a dispersional decrease after unfolding will result in a composite magnetization as well, at least if a pre-folding component constitutes more than 50% of the total "stable remanence". A crucial question in this connection is, in fact, what kind of a dispersion one is aiming at in a particular case, a problem which is clearly tied up with the actual magnetization process(es) involved. Hence, one should be very careful in drawing hasty conclusions based on straightforward correction for tectonism.

CONCLUSION

There seems to be little doubt that the present results from the Lisbon volcanics add further difficulties to palaeomagnetic research. It appears now satisfactorily documented that the primary magnetic record, not only of many older rocks, but of fairly recent ones as well, may be strongly modified by natural processes. In fact, a proper interpretation of the palaeomagnetic record of many rocks turns out to be a complicated jigsaw puzzle. Because of these difficulties some of the traditional applications of fossil magnetism are likely to be less prominent. On the other hand, the aspect of having a more or less continuous "long time" geomagnetic record even in a small rock unit opens enormous future possibilities in studies of geodynamics.

The most prominent conclusion derived from this study is that the postulated rotation of the Iberian peninsula appears to be of either Late- or post-Eocene origin; i.e. of more recent age than previously suggested. Also, a reconsideration of available marine geological evidence from the Bay of Biscay is compatible with this conclusion in that an early Oligocene age of the associated deep-sea crust can be inferred. A comprehensive discussion of the data considered (including postrotation tectonics in the area) is given elsewhere (Storetvedt, 1972). This new inference on the crustal genesis in the Bay of Biscay has a more wide-reaching aspect in that it appears to tie in very well with the time of postulated origin of the Western Mediterranean basin (Le Pichon et al., 1971) and with the climax of the Alpine orogeny.

Another interesting result is that a rapid polar shift seems to have occurred relatively soon after the rotation event. The age of this polar migration may be Late Oligocene/Early Miocene. This may well be the only polar shift in the Caenozoic of Europe. Most likely, the opening of the Bay of Biscay represents, directly or indirectly, a fragment in a world-wide sea-floor spreading episode with which a relative polar migration (possibly of global significance) is likely to be connected.

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